ASSESSMENT REMIT

Name			
ID#			
Semester	Fall 2026	**Section**	
School	School of Engineering, Architecture and Interior Design (EAID)		
Program	Bachelor of Science In Computer Science		
Major	Computer Science		
Course Code	**CSCI 310**		
Course Title	Principles of Programming Languages		
Level	☐ Remedial ☐ Bridging ☑ Undergraduate ☐ Graduate		
Assessment Title	Group Project – Build and Publish a Website or Mobile Application Using AI		
Weighting	20 Marks		
Instructor	Dr. Maher Alrahhal		
Hand Out Date	29/10/2026		
Due Date & Time	Date:24/12/2026	Time: 9:00 – 10:59 AM	
Feedback Post Date	31/12/2026		
Assessment Mode	☐ Individual ☑ Group ☐ Closed Book ☑ Open Book		
Assignment Length *(for assignment-based assessments)*	Five-page report, excluding cover page and references		
Submission Mode	☐ Online by Email ☑ Online through LMS | TurnItIn ☑ Hard Copy		
Document Type	☐ Words ☑ PDF ☐ Artwork ☑ Software ☐ Other		

GENERAL REQUIREMENTS (GROUP ASSIGNMENT)

- Each group must consist of 3–4 students.
- Develop a working website or mobile application that solves a real-world problem.
- Use VS Code with Codex or Claude, or Google Antigravity, for AI-assisted development.
- Implement at least four functional pages or screens.
- Demonstrate at least two programming paradigms in the application.
- Develop a working mini-interpreter that accepts and evaluates textual expressions.
- Test all major application features and document the results.
- Upload the complete source code and documentation to GitHub.
- Publish the website or provide an installable mobile application and demonstration.
- Submit a five-page project report, GitHub repository link and working application link where applicable.
- All group members must participate in the final demonstration.
- Students must understand, test and explain AI-generated code.
- Follow university academic integrity policies and do not publish passwords, API keys or private information.

ASSESSMENT:

GROUP PROJECT: BUILD AND PUBLISH A WEBSITE OR MOBILE APPLICATION USING AI

Group Size: 3–4 Students

Project Objective

Design, develop, test and publish a working website or mobile application using AI-assisted programming.

The application must solve a real-world problem and demonstrate programming-language concepts covered in CSCI 310.

Task 1. Project Idea, Requirements and Design (2 Marks)

Choose a real-world problem and identify:

a. The problem your application will solve.

b. The target users.

c. At least five functional requirements.

d. The main pages or screens of the application.

Suggested projects include a Smart Academic Advisor, Student Attendance System, Student Performance Analyzer, Course Registration System, Library Management System, University Helpdesk, Study Planner or Smart Parking System.

Task 2. Develop a Working Application (4 Marks)

Build a working website or mobile application containing:

- At least four functional pages or screens.
- Forms and user input.
- Data storage and retrieval.
- Search or filtering.
- Calculations or decision-making.
- Navigation and appropriate error messages.

Use suitable programming languages and frameworks.

Task 3. AI-Assisted Development and Prompt Engineering (2 Marks)

Use at least one AI coding assistant, such as Codex, Claude or Google Antigravity.

a. Document five important development prompts.

b. Show one example of improving an earlier prompt.

c. Explain one problem found in AI-generated code and how it was corrected.

Task 4. Programming Paradigms (3 Marks)

Identify and explain at least two programming paradigms used in your application.

Possible paradigms include:

- Imperative programming.
- Functional programming.
- Object-oriented programming.
- Logic programming.

For each selected paradigm, show relevant code and explain why it is suitable for your application.

Task 5. Develop a Mini-Interpreter (4 Marks)

Develop a small interpreter that accepts and evaluates expressions written in a defined mini-language.

For example:

```
AVERAGE(80,70,90) → 80

PASS(65) → True

ELIGIBLE(45,TRUE) → True
```

Your mini-interpreter must:

a. Accept a textual expression entered by the user.

b. Recognise supported commands and arguments.

c. Evaluate the expression and display the result.

d. Handle invalid expressions with appropriate error messages.

Students must explain how their interpreter processes input and produces output.

Task 6. GitHub, Publication and Project Report (3 Marks)

Create a GitHub repository containing the complete source code, README, screenshots, important AI prompts and testing results.

Publish the working website using a suitable hosting platform, or provide an installable mobile application and demonstration.

Submit a five-page report containing:

- Page 1: Introduction, problem and objectives.
- Page 2: Development tools and implementation.
- Page 3: AI-assisted coding and prompts.
- Page 4: CLO3, CLO4 and CLO5 achievement.
- Page 5: Testing, group contributions and conclusion.

Task 7. Final Presentation and Demonstration (2 Marks)

Each group must deliver a 10–12-minute presentation demonstrating:

a. The working application.

b. AI-assisted development and example prompts.

c. Two programming paradigms.

d. The working mini-interpreter.

e. GitHub repository and published application.

All group members must participate and answer questions about their contributions.

ANSWER GUIDELINES:

Students are expected to:

- Complete all seven project tasks.
- Develop a functional website or mobile application.
- Explain the programming concepts and paradigms used in their code.
- Provide evidence of AI-assisted development and prompt improvement.
- Implement and demonstrate a working mini-interpreter.
- Test the application and document errors and corrections.
- Submit complete source code, a GitHub repository and a five-page report.
- Provide the published application link where applicable.
- Demonstrate the application and explain its main features.
- Ensure all group members contribute to the project.

Presentation Requirements

- Include a cover page with all group members' names and Student IDs.
- Organise the report according to the five-page structure.
- Include screenshots, relevant code examples and testing results.
- Provide GitHub and application links.
- Use APA referencing where external sources are used.
- Clearly identify each group member's contribution.
- Participate in the final project demonstration.

REFERENCING STYLE:

Students should use the American Psychological Association style (APA) as per the following guidelines: https://apastyle.apa.org/style-grammar-guidelines/references/examples

PLAGIARISM & GENERATIVE AI (FOR ASSIGNMENT BASED ASSESSMENTS):

Amity University Dubai is committed to a policy of honesty in all its operations, especially those pertaining to academic matters. It is your responsibility to use appropriate references and apply correct referencing practices. Unless explicitly stated otherwise, the use of generative AI to write an assignment is not permitted. Submission should be yours and should not be in breach of AMUD's Academic Integrity and Academic Misconduct Policy.

LATE SUBMISSIONS (FOR ASSIGNMENT BASED ASSESSMENTS):

You are advised to strictly adhere to the deadline. While Amity University Dubai might consider extenuating circumstances to support students who are experiencing unforeseen issues impacting their ability to complete assessments on time, any anticipated submission delay must be communicated in advance to the instructor who will consult with the Program Leader and the Dean before a permission is granted. All permissions must be based on valid reasons backed up by concrete evidence. For more details, you are advised to consult the latest version of AMUD's Academic Integrity and Academic Misconduct Policy.

FEEDBACK PLAN:

Formative feedback will be provided during regular class sessions and office hours to support students in selecting their project ideas, developing applications, using AI coding assistants and implementing programming paradigms and mini-interpreters.
Students will receive guidance on code testing, debugging, GitHub publication, report preparation and group collaboration.
Summative written feedback will be provided by **31 December 2026**, based on the submitted application, source code, mini-interpreter, report and final demonstration.

ASSESSED COURSE LEARNING OUTCOMES:	RELATED QUESTIONS	MAX SCORE	MARK
CLO3: Solve problems using a range of programming paradigms and communicate the effectiveness of each paradigm for a particular problem, working in teams.	Tasks 1–2	**6**	
CLO4: Evaluate and communicate the principles of an imperative, functional, object-oriented, or logic-oriented programming language, working in teams.	Tasks 3–4, 6–7	**10**	
CLO5: Generate program interpreters for a given program code.	Task 5	**4**	
	TOTAL	**20**	

EMBEDDED PLO DIRECT ASSESSMENT QUESTIONS? ☑ **YES** ☐ **NO**
[IF YES, COMPLETE THE TABLE BELOW]

PLO	EMBEDDED QUESTIONS
PLO1: Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.	Tasks 1–2: Problem identification and application development; Task 5: Mini-interpreter implementation.
PLO3: Communicate effectively in a variety of professional contexts.	Tasks 3–4: Prompt documentation and paradigm explanation; Tasks

	6–7: Project report and presentation.
PLO5: Function effectively as a member or leader of a team, demonstrating leadership, collaboration and project management in achieving objectives related to the discipline.	Tasks 1–4 and 6–7: Collaborative planning, implementation, evaluation, documentation and demonstration.

SCORING RUBRIC			LEVEL OF ACHIEVEMENT			
	DESCRIPTOR	**WEIGHT**	**4 [87%-100%]**	**3 [73%-86%]**	**2 [60%-72%]**	**1 [0%-59%]**
1	Project Idea and Design (Task 1)	2	Clear problem, users, requirements and design.	Mostly clear with minor gaps.	Partially defined requirements.	Unclear or missing design.
2	Working Application (Task 2)	4	All required features work correctly with effective navigation.	Most features work with minor errors.	Some features work; major gaps remain.	Application incomplete or non-functional.
3	AI-Assisted Development (Task 3)	2	Clear prompts, improvements and documented debugging.	Mostly complete evidence with minor gaps.	Limited prompts or explanations.	Missing or inadequate evidence.
4	Programming Paradigms (Task 4)	3	Correctly identifies, demonstrates and evaluates two paradigms.	Mostly accurate explanations and examples.	Partial identification or weak explanations.	Incorrect or missing explanations.
5	Mini-Interpreter (Task 5)	4	Correctly parses and evaluates expressions and handles invalid input.	Mostly functional with minor errors.	Limited commands or incomplete evaluation.	Missing or non-functional interpreter.
6	GitHub, Publication and Report (Task 6)	3	Complete repository, accessible application and well-organised report.	Mostly complete submission with minor gaps.	Incomplete repository, publication or report.	Major submission components missing.
7	Presentation and Demonstration (Task 7)	2	Clear demonstration and meaningful participation by all members.	Good demonstration with minor gaps.	Partial demonstration or uneven participation.	Missing or ineffective demonstration.

Student Declaration

By signing this cover page, you acknowledge the instructions above (general requirements) and accept all responsibilities resulting from breaching them as per the related University policies. I hereby declare that it is my original work and I have duly acknowledged and referenced the source of any quotes, thoughts or ideas incorporated in the work.

Student Signature _____ **Date** _____

To be filled by faculty

FEEDBACK FROM FACULTY:

Faculty Name and Signature **Date**